

Exam Revision 2 — Intractability, Heuristics, Computability, AI

VCE Algorithmics (HESS) · Unit 4 AOS2/AOS3 drills · answers at the back

A — Classes and reductions

- A1. Define P, NP, and NP-complete in one sentence each.
- A2. Explain the direction of a reduction: if A reduces to B and A is NP-complete, what follows about B ? What does *not* follow?
- A3. Give the decision form of: “schedule the fewest exam sessions such that no student has two exams in one session.” Which studied problem does it resemble?
- A4. Why does the $O(nW)$ knapsack DP not put knapsack in P?

B — Advanced design and heuristics

- B1. Name the two properties a problem needs for dynamic programming, with a one-line test for each.
- B2. Backtracking vs brute force: what changes, what doesn’t?
- B3. For 2-opt hill climbing on TSP, define representation, neighbourhood, objective — then state the termination condition and its weakness.
- B4. In simulated annealing, what behaviour does $e^{-\Delta/T}$ produce (a) early, (b) late? Why keep a separate “best-so-far”?
- B5. A heuristic returns a tour of length 108 where the optimum is unknown. Give two honest ways to evidence its quality.

C — Computability

- C1. Components of a Turing machine; which one is “the program”?
- C2. Church–Turing thesis — statement, and why unprovable.
- C3. Reproduce the Halting Problem proof in five labelled moves.
- C4. Classify with one reason: (a) “does P halt on x within 10^9 steps?” (b) “does P halt on x ?” (c) “best TSP tour on 60 cities.”

D — AI and ethics

- D1. The Turing Test replaces which question, with what kind of criterion?
- D2. Strong vs weak AI; which does the Chinese Room attack?
- D3. State the systems reply and Searle’s counter in two sentences.
- D4. SVM objective; why does a wide margin help generalisation?
- D5. Sketch the higher-dimensions trick for points inside/outside a circle.
- D6. Overfitting vs underfitting: the error signature of each.
- D7. Ethics paragraph structure (four moves), applied to: a résumé-screening model trained on past hires.

Answers (abbreviated)

A1 P: decision problems solvable in polynomial time. NP: “yes” certificates checkable in polynomial time. NP-complete: in NP and every NP problem reduces to it. **A2** B is at least as hard as A (NP-hard); it does *not* follow that B is in NP, nor that B is exactly as hard. **A3** “Can the exams be scheduled in $\leq k$ sessions with no clash?” — graph colouring. **A4** W is exponential in the number of digits used to write it, so $O(nW)$ is pseudo-polynomial in input length.

B1 Overlapping subproblems (same subproblem appears in multiple branches of the call tree) and optimal substructure (optimal whole composed of optimal parts). **B2** Backtracking abandons partial solutions that already violate constraints — typical case improves, worst case remains exponential. **B3** Permutation of cities; all tours obtainable by reversing one segment; total length; terminate when no neighbour improves — that is a local optimum, possibly far from global. **B4** (a) frequent uphill acceptance (exploration), (b) near-zero uphill acceptance (convergence); the walk can wander off the best solution seen, so record it. **B5** Compare with exact answers on small instances; compare against a lower bound; report spread across restarts.

C1 Tape, head, finite states, transition table — the table is the program. **C2** Everything computable by a definite procedure is computable by a Turing machine; “definite procedure” is informal, so no proof is possible — only counterexamples could refute it. **C3** Assume decider H ; build $D(P)$ that loops iff $H(P, P)$ says halts; run $D(D)$; both possible answers make H wrong; contradiction — no H . **C4** (a) decidable — simulate 10^9 steps (slow but finite); (b) undecidable — Halting Problem; (c) decidable but intractable — NP-hard optimisation, finite search space.

D1 “Can machines think?” — replaced by a behavioural/operational criterion (indistinguishability in conversation). **D2** Weak: simulate intelligence; strong: literally understand; Chinese Room attacks strong AI. **D3** Systems reply: the room-as-a-whole understands even if the person doesn’t. Counter: memorise the rulebook — the whole system is now the person, who still doesn’t understand. **D4** Maximise the margin to the nearest points (support vectors); a wide margin tolerates displacement of new samples without misclassification. **D5** Add $z = x^2 + y^2$; inside-points sit low, outside high; a horizontal plane separates. **D6** Overfit: train error low, new-data error high. Underfit: both high. **D7** Stakeholder: applicants from under-represented groups; harm: systematic exclusion; mechanism: historical hiring bias encoded in labels; mitigation: audit outcomes for disparate impact / rebalance data / human review.